Environmental Analysis and Ecology
HSLU, Semester 4

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Week 1

1 TODO

Carrying capacity in general systems:
The maximum level of output or performance that a system can sustain over time without experiencing degradation or failure.

Carrying capacity in population:
Maximum population size of a species that an environment can sustain indefinitely, given the available resources and environmental conditions.

Week 2

2 Systems

2.1 Characteristics

- Input and Output
- Set of components
- Rules and relationships between components
- System's boundaries

2.2 System's loops

2.2.1 Feedback loops

Feedback loops are the system's response to changes
Two types of loops:

- Positive feedback loops (Reinforcing):
  Causes systems to change further in same direction
- Negative feedback loops (Balancing):
  Causes systems to change in opposite direction from which it is moving

2.2.2 System's responses

Time delays Complex systems often show time delays between input of a feedback stimulus and the response to it **(oscillating graph)**
Ex: Population growth or global climate change

Synergistic interaction System effects can be amplified through synergistic interaction

2.3 Casual loop diagram (CLDs)

CLDs are maps showing casual links with arrows from a cause to an effect and it allows the identification of polarity of feedback loops (positive or negative)

2.4 Reference Behavior Pattern (RBP)

Example If A increases, B will increase (↑↑)

- Arrows show same direction → assign (-)
- Variables can also be decreasing in the same direction (↓↓) → also assign (-)

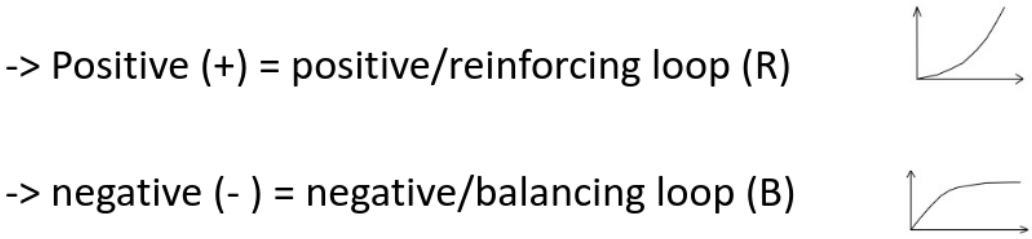


Figure 1: Reference Behavior Pattern

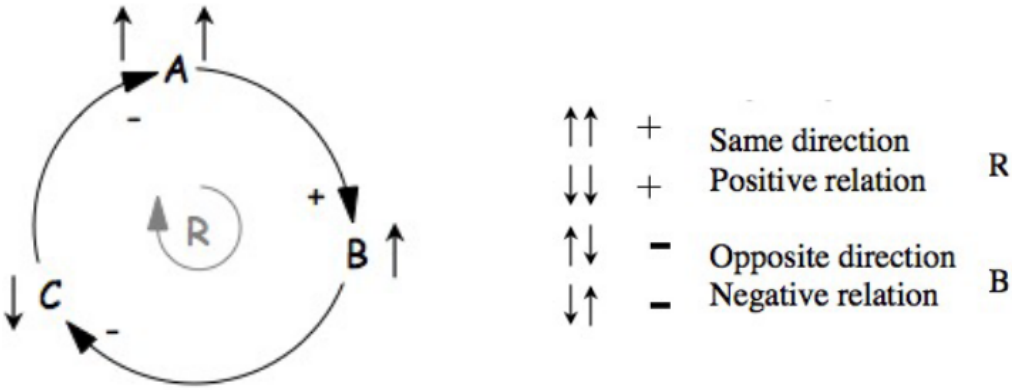


Figure 2: RBP Example

If B increases, C decreases (↑↓)

- Arrows show opposite directions → assign (-)

If C decreases, A increases (↓↑)

- Arrows show opposite directions → assign (-)

So:
A↑ → B↑ → C↓ → A↑↑ (Reinforcing Loop)
A↓ → B↓ → C↑ → A↓↓ (Reinforcing Loop)

3 Ecosystems

3.1 Energy flow in ecosystems

Robustness in ecosystems is the capacity of an ecosystem to withstand disturbances (e.g., drought, fire, pollution) while maintaining its structure and key functions.

It is important because it helps prevent collapse into degraded states and preserves ecosystem services like nutrient cycling, productivity, and biodiversity support.

3.2 Cycles in the biosphere

3.2.1 Nutrient cycle

TODO

3.2.2 Hydrologic cycle

TODO

3.2.3 Nitrogen cycle

- Major reservoir for the nitrogen is the atmosphere (78%)
- Nitrogen cannot be directly absorbed and used as a nutrient by plants or animals

Human alteration

- Humans add large amounts of nitric oxide (NO) into the atmosphere by burning any fuel at high temperatures (e.g. car, jet engines) → acid rain
- Nitrous oxide (N_2O) in atmosphere through commercial inorganic fertilizers → N_2O is a GHG and depletes stratospheric ozone
- Large quantities of nitrogen stored in soils and plants through destruction of forests, grasslands, and wetlands
- Humans alter nitrogen cycle in aquatic ecosystems through agricultural runoff and discharges from municipal sewage systems

3.2.4 Phosphorus cycle

- Phosphorus circulates in the soil mainly, also through water, the earth's crust, and living organisms
- As water runs over exposed phosphorus-containing rocks and dissolves phosphate
- Phosphate is mainly absorbed by plants roots

Human alteration

- Removal of large amounts of phosphate from the earth to make fertilizers
- Soil from fertilized crop fields carries large quantities of phosphates into streams, lakes and the ocean, where it stimulates the growth of algae
- Phosphate runoff from sewage water and mining waste

3.2.5 Sulphur cycle

- Most of the earths sulphur is stored underground in rocks and minerals, including sulphate salts buried under ocean sediments
- Sulphur also enters the atmosphere from several natural sources (e.g. active volcanoes, organic matter, which is broken down by anaerobic decomposers) → Deposited as acid rain

Human alteration

- Release of large amounts of sulphur dioxide into the atmosphere → Acid rain
  - Burning of coal and oil to produce electric power
  - Refining of sulphur-containing fossil fuels to make gasoline and heating oil
  - Conversion (smelting) sulphur-containing metallic mineral ores into free metals such as copper, lead, and zinc

3.2.6 Carbon cycle

- Carbon cycle is based on carbon dioxide (CO_2)
- 0.038% of the atmosphere consists of CO_2
- CO_2 is a key component of natures thermostat:
  - If too much CO_2 is removed from the atmosphere → atmosphere gets cooler
  - If too much CO_2 is added to the atmosphere → atmosphere gets warmer
- Plants and animals exchange CO_2 with the atmosphere through photosynthesis and respiration
- Carbon stored more permanently in plants returns to atmosphere as CO_2 through decomposition
- Oceans absorb and emit carbon to the atmosphere, and some of this carbon ends up accumulated at the bottom of the ocean
- Humans heavily contribute carbon to the atmosphere:
  - Extraction of fossil fuels → CO_2 emissions due to combustion

Week 3

4 Biodiversity

4.1 Definition

Biodiversity is the variety of earths species, the genes they contain, the ecosystems in which they live and the ecosystem processes (energy flow and nutrient cycling).

TODO

5 Species diversity

5.1 Population dynamics

5.1.1 Population

Group of individuals of the same species that live in the same place at the same time

5.1.2 Population factors

- Distribution
- Size
- Age structure
- Density

5.1.3 Population dynamics

Study of how population characteristics change in response to changes in environmental conditions.

Four factors that determine population size:

- Births
- Deaths
- Immigration
- Emigration

Population change is defined as:

$$\boxed{\text{population change} = (\text{births} + \text{immigration}) - (\text{deaths} + \text{emigration})}$$

Age structure can have a strong effect on how rapidly a population size increases or decreases:

- Pre-reproductive age (not mature enough to reproduce)
- Reproductive stage (capable of reproduction)
- Post-reproductive stage (too old to reproduce)

5.1.4 Growth functions

Species vary in their capacity for population growth under ideal conditions

5.1.5 Intrinsic rate of increase

Rate at which the population of a species would grow if it had unlimited resources.

Populations with high intrinsic rate of growth typically reproduce early in life, have short generation times (time between successive generations), can reproduce many times, and have many offspring each time they reproduce.

5.1.6 Carrying capacity

Intrinsic rate of increase and environmental resistance = carrying capacity.

Definition:
Maximum population of a given species that a particular habitat/area can sustain indefinitely without being degraded

- Carrying capacity of an area is not fixed
- A population with few (or no) limitations can grow exponentially
- Environmental resistance slows down growth rate as population size gets closer to carrying capacity:
  → logistic growth curve or S-shaped growth curve

5.2 Human influence

Extinction rate caused by human influence will increase to 10,000 times the background extinction within this century.

Annual extinction rate of ca. 1% per year (currently between 0.01 - 0.1%)

New species eventually evolve to take the places of the lost one.

Week 4

6 Stock and flow diagrams (SFD)

6.1 Translating CLDs into SFDs

- CLDs represent conceptual models, which depict causal mechanisms based on underlying reference behaviour patterns (RBPs)
- First step in computational modelling of CLDs is the translation of CLDs into SFDs

6.2 Stock and flow diagrams (SFDs)

- SFDs provide a richer visual language than CLDs
- Six basic elements exist in SFDs: **Stocks, Flows, Converters, Connectors, Sources, and Sinks**

6.3 Components of SFDs

6.3.1 Stocks

A stock is a "part of a system whose value at any given instant in time depends on the systems past behavior".

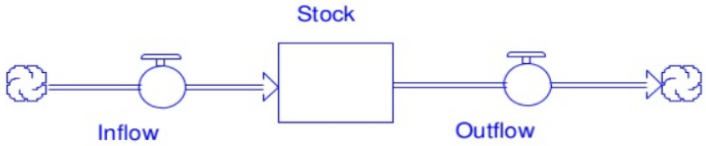
A stock is defided as:

$$\boxed{\text{Stock} = \int_{t_0}^{t_1} (\text{inflow - outflow})\, dt + \text{value right before starting}}$$

- Stocks represent anything that accumulates (normally in both directions)
- **Stocks are only affected by inflows and outflows**
- Stocks are normally tangible, countable and physical
  - But can also be non-physical (e.g. level of fear, aggression)
  - Non-tangible stocks require special attention, since differences in numerical translation might cause different system behaviours
- Stocks are usually presented by nouns

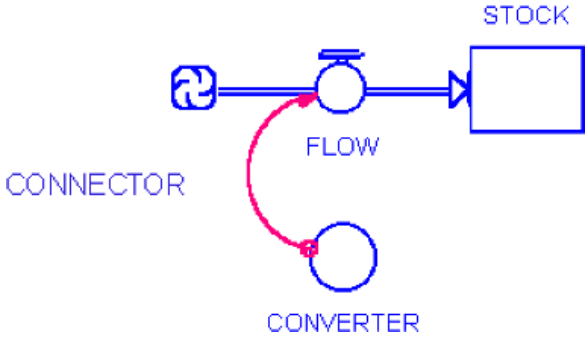
6.3.2 Flows

- Represent activities that lead to inflows and outflows to stocks (e.g. births, migration)
- Inflows cause the stock to increase
- Outflows decrease the stock
- Flows are usually presented by verbs or nouns describing an action



6.3.3 Converter

- Contain equations that generate an output value during each time interval of a simulation (e.g. birth rate, migration rate)
- Can also be used for storing constant values (e.g. initial population size)

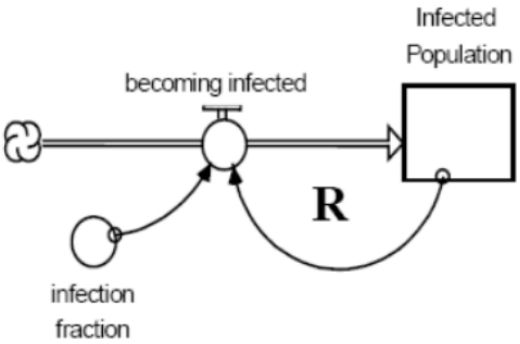


6.3.4 Connectors

- Transmit information to regulate flows
- Connectors can connect into flows or converters but never into stocks

6.3.5 Source and Sink

- Stocks that lie outside the system boundary
- Sources and sinks are shown as clouds in the diagram.
- Are used to show that a certain stock inflow is coming from another source (e.g. migration)
- Or that an outflow ends up in another stock outside the system (e.g. emigration)



Week 5

7 Climate

7.1 Weather vs. climate

Weather:

- Weather is a local, short term phenomenon
- Weather is mainly expressed by temperature, humidity, precipitation, wind speed, cloud cover
- Measured over hours or days

Climate:

- Areas general pattern of atmospheric or weather conditions
- Measured over long periods of time (ranging from decades to thousands of years)

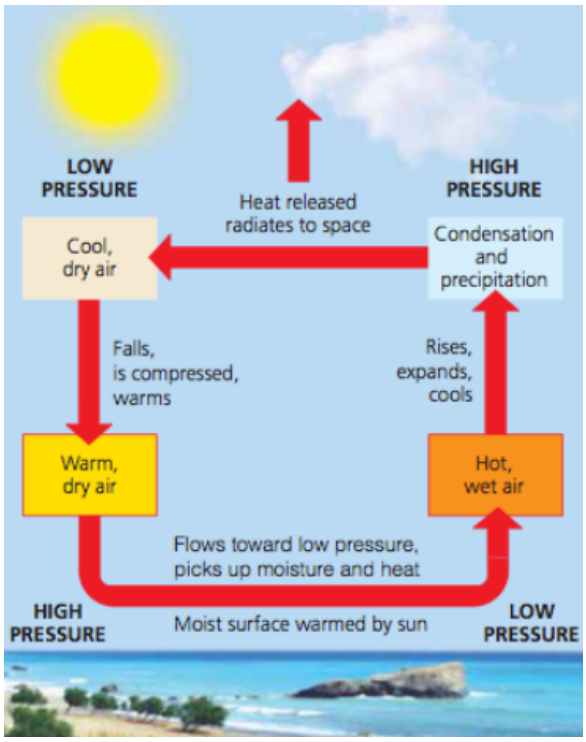
7.2 Climate factors

7.2.1 Climate zones

- Important component of earth's natural capital
- Climate variations caused by patterns of global air circulation and ocean currents

7.2.2 Factors of air circulation in lower atmosphere

- Uneven heating of earth's surface → Air is heated more at the equator
- Rotation of the earth around its axis:
  - Heated air masses rise above equator and move north and south
  - Atmosphere divided in cells, distinguished by direction of air movement
  - Differing directions of air movement are called **prevailing winds** (major surface winds that blow almost continuously and distribute air, heat, moisture and dust)
  - Prevailing winds also produce mass movements of surface water (currents)
- Properties of air, water and land:
  - Heat from the sun evaporates ocean water and transfers heat from the oceans to the atmosphere, especially near the equator
  - Evaporation of water creates cells (low and high pressure) which circulate air, heat, and moisture



7.2.3 Ocean conveyer belt (thermohaline circulation)

- Heat is also distributed when ocean water mixes vertically in shallow and deep ocean currents
- Result of differences in the seawater density
- Colder seawater (higher density) sinks and flows beneath warmer and less dense seawater
- Connected loop of deep and shallow ocean currents

- Acts like giant conveyer belt that moves heat to and from the deep sea
- Transfers warm and cold water between tropics and poles

7.2.4 Link between ocean and atmosphere

- Ocean currents are affected by winds in the atmosphere
- Heat from the ocean affects atmospheric circulation

7.2.5 Earth's surface features

- Heat is absorbed and released more slowly by water than by land:
  → Difference creates land and sea breezes → oceans and large lakes moderate weather and climates of nearby lands
- Topographic features of earths surface can create local and regional weather and climatic conditions that differ from the general climate of a region (e.g. mountains)

7.2.6 Greenhouse effect warms lower atmosphere

- Greenhouse gases allow visible light (and some infrared radiation and UV radiation) from the sun to pass through the atmosphere
- Earths surface absorbs much of this solar energy and transforms it to heat, which then rises into the lower atmosphere
- Some of this heat escapes into space, but some is absorbed by molecules of greenhouse gases → Warms the lower atmosphere and the earths surface (greenhouse effect)

8 Climate change - natural cycles

8.1 Characteristics of climate change

8.1.1 Natural variations

- Changes of earths climate are neither new nor unusual
- Atmosphere has experienced prolonged periods of global cooling and global warming
- Alternating cycles of cooling and warming are called glacial and interglacial periods

9 Natural forces

9.1 Milankovitch cycle

9.1.1 Variations

Earth's orbit
Earths orbit changes from being nearly circular to slightly elliptical (termed eccentricity)

→ Cycle has a period of around 100.000 years

→ The more elliptic the orbit becomes, the less time the earth spends near the sun

→ Less solar energy leads to cooling effect

Angle of tilt
Angle of tilt of the earths axis changes from 22,1° (22,5) to 24,5° (term obliquity)

→ Cycle has a period of 41.000 years

→ Land mass of the northern hemisphere face more/less towards the sun

→ Leads to cooling/warming effects

Direction of tilt
Direction of tilt of the axis changes (termed precession)

→ Cycle has a period of 26.000 years

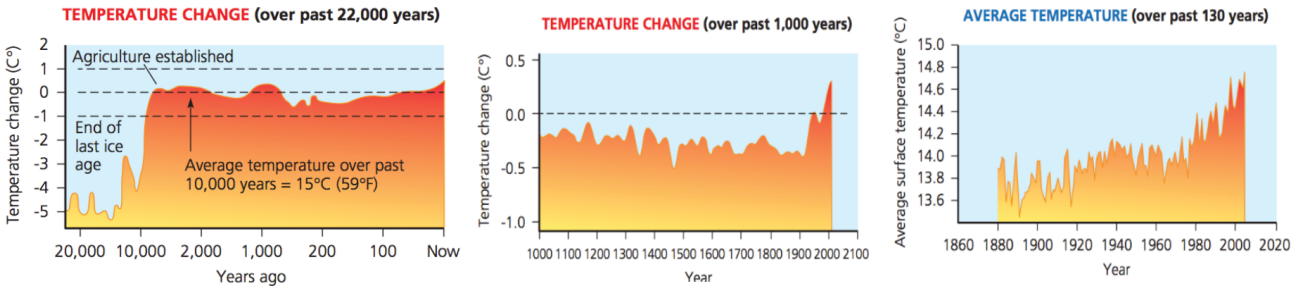
→ Causes winters and summers to be warmer or colder depending on the amount of land surface being more or less exposed to the sun

10 Anthropogenic climate change

10.1 Global warming

10.1.1 Today's climate

- For ca. 10,000 years earths climate is in an interglacial period
- It is characterized by a fairly stable climate and a fairly steady average global surface temperature
  → Allows agriculture → triggered population growth
- Average temperature of the atmosphere has remained fairly stable for the last 1,000 years
- Temperature started rising during last century



The 0 temperature change line was set based of the average temperature right after the beginning of the industrial revolution.

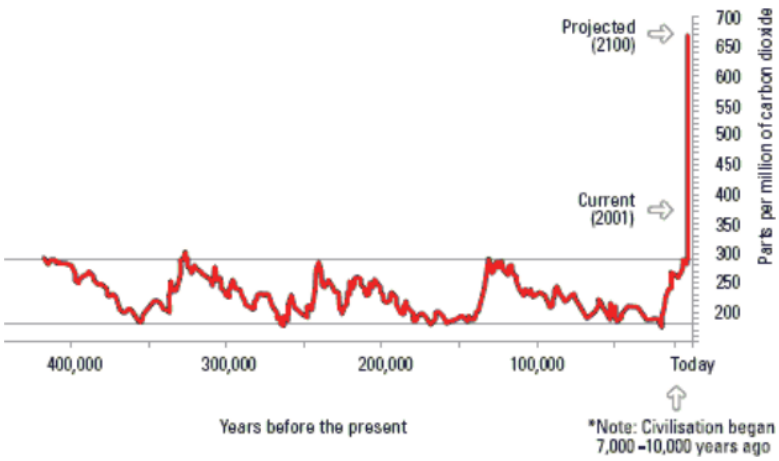
10.2 Greenhouse effect

- Life depends on natural greenhouse effect → Without natural greenhouse effect no life would exist and earth would be too cold
- Since Industrial Revolution significant increase in CO_2, CH_4, and N_2O concentration in lower atmosphere
  → Increases is mainly due to agriculture, deforestation and burning of fossil fuels
- Measurements of CO_2 and CH_4 in glacial ice cores indicate high correlation between concentration level and average global temperature

10.3 Greenhouse gas emission

- CO_2 level has risen from 280 ppm (around 275 years ago) to 399 ppm in 2014
- If CO_2 emissions continue to increase at current rate of about 3.3% a year
  → Levels will rise to 560 ppm by 2050 and to 1,390 ppm by 2100
  → Significant changes in earths climate
  → Major ecological and economic disruptions
- CO_2 levels should be prevented from exceeding 450 ppm
  → Estimated tipping point that sets into motion large-scale climate changes
- Ideally: reduction of CO_2 to 350 ppm (pre-industrial level)
  → Stabilizing earths climate

10.3.1 CO_2 emission over time



Due to the periodic changes in the Earth's orbit, from elliptical to circular and vice versa, the more elliptical the orbit is, the farther the Earth is from the Sun, the fewer plants there are, and the less CO_2 is stored.

Nowadays, the amount of CO_2 in the atmosphere is higher than ever due to industrialization.

10.3.2 Global Warming Potential (GWP)

- Describes potential impact of each greenhouse gas (GHG)
- Some GHGs are more effective at warming the atmosphere than others
- Most important characteristics of a GHG in terms of climate impact are:
  - → Energy absorption potential (preventing energy from escaping to space)
  - → Residence time of gas in atmosphere
- GWP measures total energy that gas absorbs over 100 years, compared to CO_2

Greenhouse Gase	Chemical formula	Global Waeming 20 years	Potential (Time horizon) 100 years
Carbon dioxide	CO_2	1	1
Methane	CH_4	42-70	16-26
Nitrous oxide	N_2O	280 $\longleftrightarrow$	310

10.3.3 Methane CH_4

- 60% of methane emissions occurred during last 275 years
- Extraction of fossil fuels, creating landfills, raising cattle and sheep
- CH_4 emissions have levelled off since 1990
- Are expected to rise again → melting of permafrost in the arctic tundra
- GWP: 16-26 times greater than CO_2

10.3.4 Nitrodus oxide N_2O

- Nitrous oxide levels have risen ca. 20% during last 275 years
- Increased use of nitrogen fertilizers
- GWP: 280 times greater than CO_2

10.4 Effects of a warmer atmosphere

10.4.1 Extreme weather events

- Increased incidence of extreme weather events such as heat waves, droughts, flooding
- Controversy over the question if global warming also increases frequency and intensity of tropical storms and hurricanes
- Hurricane Katrina caused loss of over 320 million big trees
  → dead and damaged trees emitted CO_2 equal to the total amount that all forest trees in the U.S. absorb in one year

10.4.2 Effects on biodiversity

- ca. 30% of the land-based plant and animal species could disappear if average global temperature change exceeds 1.5-2.5 °C
- Percentage could grow to 7% if temperature change exceeds 3.5 °C
- Hardest hit will be plant and animal species in colder climates
- Global warming could also disrupt biological clocks of birds, whales, and other migratory species
- Ecosystems will suffer disruption (e.g. coral reefs, polar seas) → species extinction

10.4.3 Effects on humans

Nutrition

- Agricultural productivity may increase in some areas and decrease in others
- Crop productivity is projected to increase slightly at middle to high latitudes
- Agricultural productivity will decrease in tropical and subtropical regions
- Flooding of rivers and droughts reduces crop production
- IPCC warns that ca. 200-600 million people could face malnutrition/starvation from climate change effects

Health

- Fewer people will die from cold weather, but heat-related deaths will increase
- Better living conditions for rapidly multiplying insects, microbes, toxic moulds
- Infectious diseases (e.g. dengue fever, yellow fever, malaria) are likely to expand their ranges
- Higher atmospheric temperatures also increases some forms of air pollution
  → Speeds up chemical reactions, which causes photochemical smog in urban areas
- WHO estimates that currently each year, climate change contributes to premature deaths of more than 150'000 people

10.4.4 Reduction of GHG emissions

4 steps for CO_2 reduction

1. Change the type of fuel (conversion of primary source)
2. Maximize the efficiency of conversion of the fuels
3. Increase the energy efficiency of the device itself
4. Change the behavior of the humans

Major GHG reduction strategies

- Improving energy efficiency to reduce fossil fuel use
- Shift from non-renewable (carbon-based) fuels to (carbon-free) renewable energy sources
- Stop cutting down tropical forests

- Effectiveness of these strategies would be enhanced by reducing population

  → Decreasing number of CO_2 emitters

  → Reducing poverty

  → Decreasing need to clear more land for crops and fuel wood

Geo-engineering

- Capturing and storage of CO_2 (and keep burning fossil fuels)
- Injection of sulphate particles into stratosphere → reflection of incoming sunlight
- Space sunshades

Week 6

11 TODO

Looks like we skipped this week

Week 7

12 Analytical tools for energy and environmental systems

12.1 Characteristics of energy systems

12.1.1 Relationship between energy use and wealth

- Most used measure for wealth = GDP
- GDP = sum of monetary value of all goods and services produced in a country/year
- GDP value may be adjusted to reflect purchasing power parity (PPP) → A dollar equivalent earned in one country may not buy as much as in another country
- GDP per capita is correlated with energy use per capita → GDP rises with energy use (per capita)

12.1.2 Quality criteria of energy resources

- Must be reliable and deliver the service that the consumer expects
- Must be available in the quantity desired and when the consumer wishes to consume it
- Resource must be available at a price that is economically affordable
- If quality criteria are not met, direct consequences are experienced

12.1.3 Long term criterion

- Another criterion for energy resource is environmental sustainability
- Problem: consumers of energy resources do not experience direct consequences if this criterion is not met

12.2 Energy systems categories

- Physical goals
- Financial goals
- Environmental goals

12.2.1 Physical goals

- Physical requirements that make it possible for the system to operate
- All systems require primary energy resources
- Any type of energy system must function efficiently, reliably, and safely

12.2.2 Financial goals

- For small-scale, privately held systems goal might be to reduce energy
- For commercial energy systems the goal is normally profit maximisation (returning profits to shareholders)

12.2.3 Environmental goals

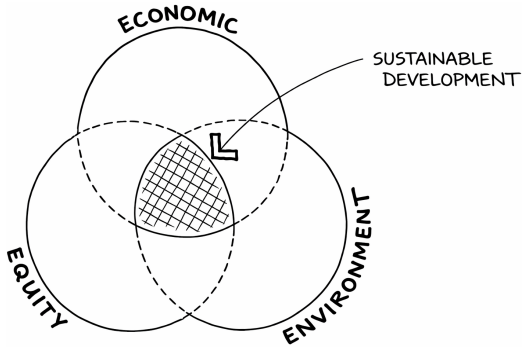
- Reduction of GHG emissions
- Reduction of air pollutants → increase of air quality
- Minimization of physical effects from extracting resources
- Minimization of impacts on physical surroundings, such as noise or vibrations
- Reduction of (thermal) pollution of water
- Avoidance of disruption of natural habitats

12.3 Objectives

- Use of existing technology options in a different way
- Conserving energy by replacing end-use technology → Primary energy source remains the same, but new technology is introduced to use energy resource more efficiently
- Conserving energy by replacing conversion technology → Same primary source of energy (e.g., fossil fuels), but conserve energy by replacing the energy conversion device with a more efficient model
- Replacing existing energy sources with alternatives

12.3.1 Overall sustainability objectives (Sustainable development)

- Environmental sustainability: protection of the environment
- Social/cultural sustainability: increasing equity (decreasing poverty)
- Economic sustainability: responsible economic development

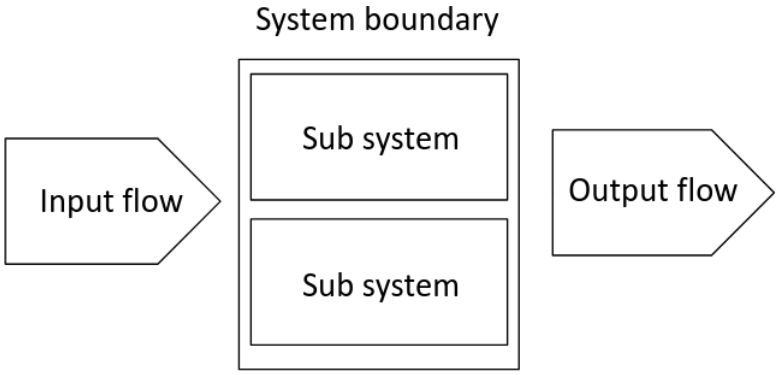


12.4 Definition of energy systems

- "A system is a set of components that function and interact in some regular way"
- Group of interacting components that work together to achieve some common purpose
- Each system has a boundary: physical or conceptual → Entities or objects that are not part of system lie outside the boundary
- Area outside the boundary is called the environment

- Energy systems have input and output flow between the system and its environment
- Inputs and outputs may be physical (e.g. raw materials) or virtual (e.g. information, data)
- Systems can consist of subsystem (e.g. electricity grid)
- Defining boundaries, input and output flows and subsystems is essential

12.4.1 Energy systems diagram



where:

- **System boundary**: defines what is included in the analysis and what is outside the system to study the relevant components and interactions
- **Input flow**: represents the resources entering the system, such as energy, materials, or information
- **Sub system**: a smaller internal part of the main system with a specific function
- **Output flow**: represents what leaves the system, such as useful energy, waste, emissions, or information

12.5 Energy return on investment (EROI)

- End-use energy derived from a particular resource needs to justify energy expended upstream in its extraction, conversion and delivery:
  - **Upstream: everything before the operational phase**
  - **Downstream: energy or money used to dismantle the energy system or to end it**
- EROI = energy delivered at the end of the life cycle / energy used in earlier stages
- EROI presents the amount of energy returned from the use of an energy source after expending energy on manufacture, installation etc.

12.6 Energy payback time (EPBT)

- EPBT expresses the years it takes for a certain technology to produce as much net energy as it takes to manufacture and dispose the technology
- Used as a measure of energy efficiency from an investment perspective
- EPBT must be much less than lifetime of technology

$$\boxed{\begin{gathered}\text{EPBT} = \frac{\text{Cumulative energy demand}}{\text{Yearly net energy generated}} \\ \text{EPBT} = \frac{E_{\text{Materials}} + E_{\text{Manufactoring}} + E_{\text{Installation}} + E_{\text{Disposal recycling}}}{\text{Energy generated} - E_{\text{O\&M}}}\end{gathered}}$$

where $E_{\text{O\&M}}$ is the energy used in operations and maintenance, which is subtracted from the energy generated

12.7 Kaya equation (Kaya identity)

- System approach for analysing factors that contribute to overall CO_2 emissions (country or global base)
- Approach includes population, level of economic activity, level of energy consumption, and carbon intensity

$$\boxed{CO_2 = P \cdot \frac{GDP}{P} \cdot \frac{E}{GDP} \cdot \frac{CO_2}{E}}$$

where:

- P: population (of a specific country)
- GDP/P: production per capita (to evaluate how wealthy the population is)
- E/GDP: energy intensity per unit of GDP (energy intensity of the population's economy / how much energy do you need to produce $1)
- CO_2/E: carbon emissions per unit of consumed energy

13 Life Cycle Assessment (LCA)

LCA is an analytical tool that assesses energy consumption/emissions within a life cycle of a system (e.g. product, energy system, material)